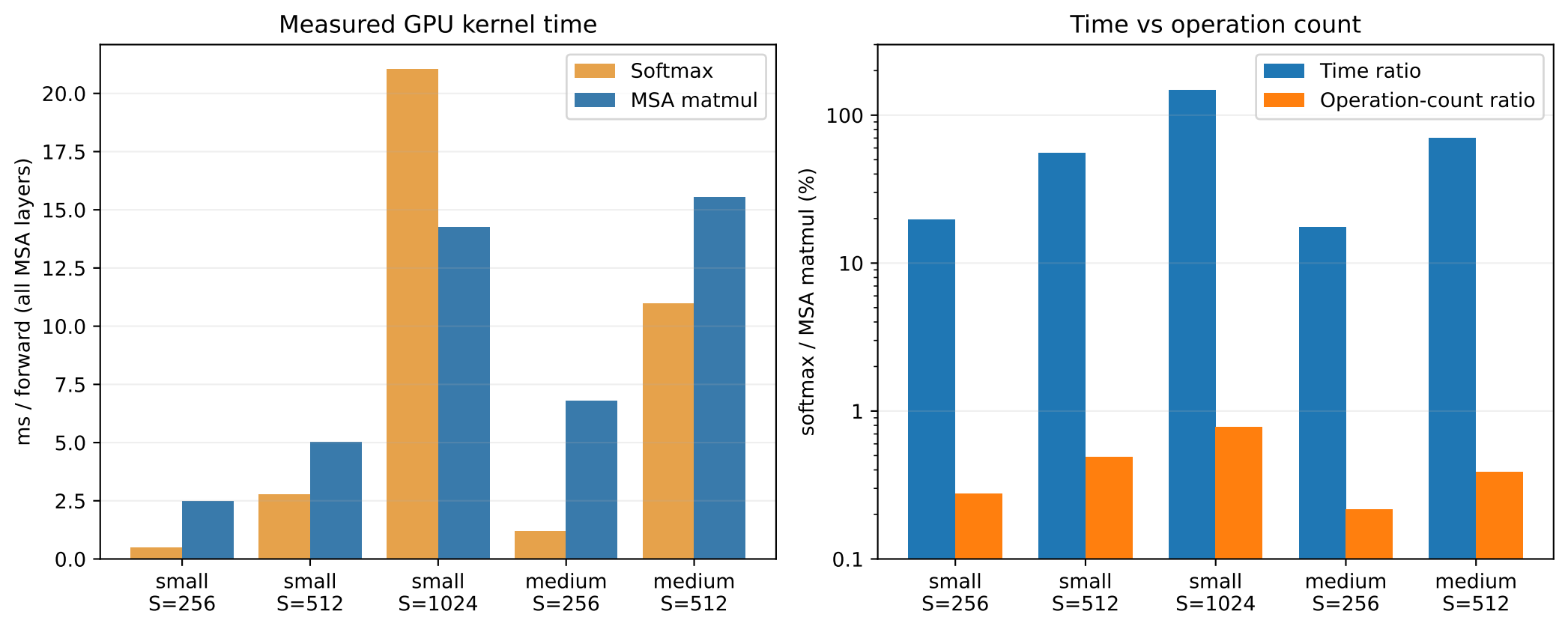
Measured GPU kernel time
ms / forward (all MSA layers)
Softmax
MSA matmul
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
small S=256
small S=512
small S=1024
medium S=256
medium S=512
Time vs operation count
softmax / MSA matmul (%)
Time ratio
Operation-count ratio
0.1
1
10
100
small S=256
small S=512
small S=1024
medium S=256
medium S=512